

Miles Franck

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SUMMARY

Software engineer with a technical foundation in Python, database systems, and systems-level programming in Rust, including concurrent and async application design. Comfortable using AI tools to support development and analysis workflows. Experienced building backend architectures, debugging complex systems, and collaborating across technical and non-technical stakeholders. Brings a security-aware development mindset from SOC analyst experience, with CompTIA Security+ in progress. Looking to grow within a collaborative engineering team building reliable, well-architected software

CORE COMPETENCIES

Software Design & Architecture • Concurrent / Async Programming • Database Design • Debugging & Troubleshooting • Secure SDLC • Agile / Scrum Collaboration • Technical Documentation

TECHNICAL SKILLS

Languages & Databases: Python, Rust, PostgreSQL

Systems & Tooling: Git, WSL, Tokio (async runtime)

Security Tools (Familiar): Shodan, Nmap, Splunk, Wireshark

Education

Bachelor of Science in Software Engineering — Allegheny College Graduated May 2025

EXPERIENCE

Summer Fellow — Carnegie Mellon University, Heinz College Summer Fellowship *2025*

- Completed coursework in networking fundamentals (TCP/IP), database systems, and information security
- Partnered with an external organization on a team project, gathering stakeholder requirements and evaluating SIEM tool capabilities against defined technical criteria (log ingestion, alerting, usability, cloud support)
- Delivered a live technical presentation of findings to an audience including CMU staff and outside industry guests

SOC Operations Analyst II — BCR Cyber *Dec 2025*

- Participated in a live-fire cyber range exercise, identifying and tracing an active attacker's presence within a simulated enterprise environment

SOC Operations Analyst I — BCR Cyber *Nov 2025*

- Applied technical troubleshooting and analytical skills within security operations workflows, including log analysis, alert triage, and structured incident documentation.

Teaching Assistant — Allegheny College *2022 – 2025*

- Provided one-on-one technical support and debugging help to students during office hours, translating complex concepts into clear, actionable guidance.
- Reviewed student code and projects for correctness, structure, and adherence to best-practice coding standards.
- Communicated across technical and non-technical audiences to align grading criteria and project expectations with faculty.

PROJECTS

Rust TCP Chat Server *2026*

- Built a multi-client TCP chat server in Rust using the Tokio asynchronous runtime, supporting concurrent client connections and real-time message broadcasting.

- Implemented a publish-subscribe messaging system using Tokio broadcast channels to efficiently distribute messages to all connected clients.
- Designed a room management system with unique join codes, user authentication, connection limits, and participant tracking.

BanjoDB *2025*

- Built a dual-language architecture separating a Rust core from a Python interface to emulate enterprise-style backend isolation.
- Implemented Argon2 password hashing with unique per-user salts as part of the application's authentication layer.

Automated Vulnerability Analysis Tool *2024*

- Built a Python tool that queries the Shodan API to enumerate internet-facing assets and cross-references results against CVE/CPE vulnerability databases.

Deepseeking-Vulns *2025*

- Developed an AI-assisted analysis workflow integrating Shodan, Ollama, and DeepSeek to process and evaluate exposed network service data.

CERTIFICATIONS

- CompTIA Security+ — CompTIA (Expected October 2026)
- SOC Operations Analyst II — BCR Cyber (Dec 2025)
- SOC Operations Analyst I — BCR Cyber (Nov 2025)
- Network Defenses 95-884 — Carnegie Mellon University, Heinz College (2025)